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CAPSTONE PROJECT

CHAPTER 1 - INTRODUCTION

Student / Group Template

Project Title	Customer Support Ticketing System: Design and Development of a Web-Based Support Request Management Application
Group Members	Freek Schaefer-Gerber; Wian Sandenbergh
Module	Software Engineering Capstone Project
Lecturer / Supervisor	To be completed
Week / Stage	Proposal and initial research
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1.1 Background of the Study

Customer support is an important part of service delivery because organisations must respond to questions, complaints and requests in a timely and consistent manner. In smaller support environments, enquiries may arrive through different channels such as email, telephone or messaging applications. Without a common tracking process, staff can lose context, duplicate work or fail to communicate progress. Ticketing systems offer a structured approach by assigning each request a reference number, a responsible agent and a documented status. This capstone project proposes a web-based customer support ticketing system intended as a practical prototype for small service teams. No particular organisation has yet been selected as a deployment client; user needs and workflow assumptions will be refined through requirements analysis.

1.2 Problem Statement

A support team relying on disconnected messages or manual spreadsheets may have difficulty tracking which customer issues are open, who is responsible, what actions have been taken and whether a request has been resolved. Customers may need to repeat information or ask for updates, while administrators lack a reliable overview of workload and resolution progress. The problem addressed by this project is the absence of a central, accessible and auditable workflow for recording and managing customer support requests in the proposed small-team setting. The extent of these problems will be investigated during requirements gathering rather than assumed to have been measured.

1.3 Research Aim

To design, develop and evaluate a web-based customer support ticketing prototype that centralises customer requests, supports agent assignment and status tracking, and improves the visibility and traceability of support workflows.

1.4 Research Objectives

1. Analyse the support workflow and document functional and non-functional requirements for customers, support agents and administrators.
2. Design the system architecture, interface prototypes and normalised relational database, including an entity-relationship diagram.
3. Develop a working application supporting authentication, ticket creation, categorisation, assignment, status changes and comments.
4. Implement access controls, validation and a basic dashboard to support secure and organised ticket handling.
5. Test the prototype using defined test cases and evaluate the implemented functions against the documented requirements.

1.5 Research Questions

1. What functions and quality requirements are needed for a small-team customer support ticketing workflow?
2. How should the application interface, architecture and database be designed to support reliable ticket management?
3. To what extent can the developed prototype support the full ticket lifecycle from submission to closure?
4. How can access control, validation and dashboard summaries support the different user roles?
5. Does the prototype satisfy its specified requirements when assessed through systematic functional tests?

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1.6 Proposed Software Solution

The proposed solution is a responsive web application with three primary roles: customer, support agent and administrator. A customer creates an account, submits a ticket with a subject, description and category, and follows its progress. Agents see relevant tickets, add responses and update statuses. Administrators assign tickets, manage categories and review ticket counts by status or priority. Each ticket receives a unique identifier and stores creation and update timestamps. A relational database will maintain user accounts, tickets, comments, categories and status history. The technology stack will be confirmed in the requirements and design phase; React, ASP.NET Core and SQL Server are candidate technologies rather than completed implementation decisions.

1.7 Scope

In scope are the design and implementation of a demonstrable prototype; user authentication and authorisation; customer ticket submission and viewing; ticket categories and priorities; agent assignment; status transitions; customer-agent comments; search/filter functions; administrative summary metrics; database design; and testing evidence. Out of scope for the first version are a production deployment commitment, integrations with external CRM platforms, payment processing, AI-assisted responses, telephony integration and complex SLA escalation rules. The project will use fictional or consented test data and will not require collecting real customer complaints.

1.8 Significance of the Study

The project will demonstrate how established software engineering practices can be applied to a practical service-management problem. For customers, the proposed system could provide a clearer way to submit requests and check progress. For support staff, a central record of tickets and responses could reduce ambiguity about ownership and next actions. For administrators, dashboard summaries could improve workload visibility. Academically, the project provides evidence of requirements analysis, system and database design, application development, teamwork, version control and structured testing. These benefits are expected outcomes to be assessed, not established results.

1.9 Chapter Summary

This chapter introduced the proposed customer support ticketing system, the support-management problem it intends to address, the research aim and objectives, the guiding questions and the planned scope. The next chapter will review relevant literature and existing ticketing solutions to identify useful design practices, limitations and a defensible research gap. Later chapters will document requirements, system design, implementation, testing and evaluation.